FINITE PERTURBATIONS CAN DESTROY FOURIER INTERPOLATION

ØISTEIN SØVIK

> For example, does moving (but not removing) finitely many interpolation points preserve the interpolation property?

The unnumbered question after Conjecture 7.5, Section 7, preprint p. 96, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, doi:10.4007/annals.2022.196.3.3, Ann. Math. 196 (2022), 983–1082. The interpolation data at each radius are $f, f', \widehat{f}, \widehat{f}'$; the canonical radii are $r_n = \sqrt{2n}$ in dimensions 8 and 24.

Answered: no. Evidence: Sketch.

No. In each of those two systems a collision-free finite move makes the data fail to determine the radial Schwartz function.

Proof. Put $n_0 = 1$ in dimension 8 and $n_0 = 2$ in dimension 24. The source's interpolation basis has radial Schwartz functions a_j, b_j with cardinal value and derivative data at the canonical radii; their Fourier transforms have zero canonical data. They are real, and radial Fourier inversion gives $\widehat{\widehat{h}} = h$. For a fixed sign $\epsilon \in \{1, -1\}$, set $u_j = a_j + \epsilon\widehat{a}_j$ and $v_j = b_j + \epsilon\widehat{b}_j$. Then $\widehat{u}_j = \epsilon u_j$, $\widehat{v}_j = \epsilon v_j$, and the two value and derivative rows at each canonical radius form the identity matrix.

Fix n and define $W(r) = u_n(r)v_n'(r) - u_n'(r)v_n(r)$. Cardinality gives $W(r_n) = 1$ and $W(r_m) = 0$ for $m \neq n$. For the centered dilation operator $D = x \cdot \nabla + d/2$, Fourier transformation gives $\widehat{Dh} = -D\widehat{h}$. Parseval's identity and the opposite Fourier parities imply $\int_{\mathbb{R}^d} u_n Dv_n = 0$. In radial coordinates, integration by parts converts this to $\int_0^\infty r^d W(r)\,\mathrm{d}r = 0$. Since W is positive near r_n, it is negative at some positive noncanonical radius t.

Choose M with $t < r_{M+1}$. On the finite span of u_j, v_j for $n_0 \le j \le M$, the canonical evaluation matrix is the identity. Replace r_n by t; expansion along all unchanged rows makes the determinant $W(t) < 0$. Sorting the radii only permutes blocks of two rows and does not change the sign. The straight path from the canonical ordered nodes to these ordered nodes stays positive, distinct, and below r_{M+1}. Continuity therefore gives an intermediate moved system with singular evaluation matrix. A nonzero real kernel vector gives a nonzero radial Schwartz function whose values and derivatives vanish at every moved node and every unchanged tail node; Fourier parity makes the Fourier data vanish as well. All four data are moved together, without collisions. □

AI assistance was used in adapting and checking the argument and the exposition.

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